

MOSTOFA HABIB FARDIN

Lecturer, Department of Industrial and Production Engineering, BUET

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ABOUT ME

Lecturer in the Department of Industrial and Production Engineering at BUET, where I earned my B.Sc. (CGPA 3.88). My work bridges rigorous quantitative research with practical industrial problems, focused on supply chain simulation, machine learning forecasting, and information systems in operations. I teach undergraduate production and manufacturing courses while continuing my research in operations and supply chain management.

EDUCATION

B.Sc. in Industrial and Production Engineering

June 2026

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

CGPA: 3.88 / 4.00

Thesis: Taming the Bullwhip Under Geopolitical Uncertainty: A Multi-Echelon Simulation of Machine Learning Forecasting and Blockchain-Enabled Information Visibility

Supervisor: Dr. Sultana Parveen, Professor, Department of IPE, BUET

Higher Secondary Certificate (HSC)

2020

Notre Dame College, Dhaka, Bangladesh

GPA: 5.00 / 5.00

ACADEMIC EXPERIENCE

Lecturer | Department of IPE, BUET

July 2026 – Present

(Lecturer, Part-time, June – July 2026)

- Teaching undergraduate courses:
 - IPE 331: Production Processes, IPE 332: Production Processes Sessional, IPE 204: Engineering Graphics & Introduction to CAD Sessional, IPE 432: Machine Tools Sessional
- Course and lab-session planning, technical instruction in SolidWorks, and student mentorship.

RESEARCH

Undergraduate Thesis

June 2026

Taming the Bullwhip Under Geopolitical Uncertainty: A Multi-Echelon Simulation of Machine Learning Forecasting and Blockchain-Enabled Information Visibility

Co-author: Sheikh Mahir Abrar • Supervisor: Dr. Sultana Parveen, Professor, BUET • **Manuscript in preparation**

- Built a five-echelon discrete-event supply chain simulator across 5,740 simulation days, jointly modelling ML forecasting, geopolitical lead-time disruption, and blockchain visibility in a single integrated framework.
- Benchmarked six forecasting models across 1,000+ simulation runs, validated with t-tests and Cohen's d effect-size analysis.

- Found that the Temporal Fusion Transformer suppressed the bullwhip effect most effectively, despite lower point-forecast accuracy than other ML models.
- Observed that ARIMA became structurally unstable under geopolitical disruption (coefficient of variation exceeding 70% and two numerical overflows), a failure mode not previously documented in the supply chain simulation literature.
- Showed that pipeline-only blockchain visibility performed statistically on par with full transparency, suggesting firms need not share sensitive data to gain the core benefits of blockchain adoption.
- Confirmed geopolitical disruption as the single largest driver of bullwhip amplification, outweighing forecasting method and information visibility combined.

Research interests: supply chain resilience, machine learning, information systems in operations, operations management.

INDUSTRIAL EXPERIENCE

Nextern Intern | Grameenphone Ltd.

June 2026

Consumer Mobility Business, Product Division, Dhaka, Bangladesh

- Supported regulatory reporting operations by collecting, sorting, and consolidating data from multiple internal teams for senior review, within a large-scale telecommunications product environment.

Industrial Attachment | RANCON Industrial Park

Nov – Dec 2025

RANCON Auto Industries Ltd. & RANCON Motorbikes Ltd.

- Worked in the Production Line and Quality Control Department on time study, tool listing, SOP upgrades, and Rack ID system implementation.
- Identified production bottlenecks and safety hazards, and presented recommendations directly to the Executive Director.
- Contributed to 5S, Kaizen, and safety-enhancement initiatives, aligning operations with Japanese OEM standards.

PROJECTS

HydroPedal: A Pedal-Powered Water Pump

- Designed a low-cost pedal-powered water pump for off-grid rural households as an alternative to motorised pumps and tube-wells.
- Modelled and validated the chain-sprocket drive system in SolidWorks and ANSYS, optimising gear ratio and frame design for local manufacturability.
- Achieved a flow rate of 25–30 L/min at 20 ft head with zero fuel or electricity cost.

Smart IoT-Based Ammonia and Ambient Condition Monitoring System

- Built an IoT system to monitor ammonia concentration and ambient conditions for fertiliser-plant safety applications.
- Integrated MQ137 and DHT22 sensors with ESP-01S for real-time data logging, with automated fan and buzzer alerts at threshold levels.
- Developed a web dashboard for data visualisation and safety-compliance reporting.

SKILLS

Simulation & Modelling: SolidWorks, CATIA, AutoCAD, ANSYS

Programming & Machine Learning: Python (XGBoost, LightGBM, TFT, ARIMA, Pandas), MATLAB, C

Operations & Quality: Lean, Kaizen, 5S, Six Sigma (White Belt), Time Study

Teaching & Communication: Curriculum and lab planning, technical instruction, mentorship, LaTeX

Soft Skills: Problem-solving, Teamwork, Critical Thinking, Time Management, Coordination

AWARDS

• Dean's List Award, BUET	2022, 2023
• IRC WASH Water Hackathon, Finalist	2025
• ISCEA Global Supply Chain Case Competition, 60% Scholarship	2025
• HULT Prize at BUET, 1st Runner-up	2022
• Bizzmania (BUET EDC), Champion	2022
• Perfect Attendance Certificate, Notre Dame College	2020

REFERENCES

Dr. Sultana Parveen

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Bangladesh University of Engineering and Technology

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Dr. Abdullahil Azeem

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